

Hand calculations, step by step: from the hard-disk sound speed to dissipation and lost information

Cowork note for Chris — HardDisks / hspist3

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How to use this

Every section is one link in the chain we discussed: geometry of route A $\rightarrow$ kinetic numbers (mean free path, Knudsen) $\rightarrow$ what “ideal gas” means for the sound speed $\rightarrow$ how c_s is measured with the divider $\rightarrow$ the energy ledger of a piston push $\rightarrow$ quasi-static work and dissipation $\rightarrow$ Jarzynski $\rightarrow$ linear response $\rightarrow$ nostalgia. Each derivation is written so that you can redo it on paper; the coloured boxes give the numbers to plug in and the checks that catch mistakes.

Units used throughout. Simulation units $k_B T = m = \sigma = 1$, where $\sigma = 2r$ is the disk *diameter* ($r = 0.5$). The unit of speed is $\sqrt{k_B T/m}$, the unit of time is $\sigma \sqrt{m/k_B T}$ (“ σ -time”). The code’s pixel time is $24\times$ finer: $t_\sigma = t_{\text{px}}/24$. Packing fraction $\eta = N\pi r^2/A$. Two-dimensional gas, so each particle carries $\langle \frac{1}{2}mv^2 \rangle = k_B T$ (two quadratic degrees of freedom, $\frac{1}{2}k_B T$ each).

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1 Geometry of route A: why $\eta = 0.02$ is not “two particles”

Route A keeps the number of disks fixed and changes the box. Per run: $N = 100$ disks, 50 per compartment, radius $r = 0.5$, height $H = 10$, two compartments of length L_0 each, separated by the divider. The packing fraction is the disk area over the total gas area:

$$\eta = \frac{N\pi r^2}{2L_0 H} = \frac{100 \cdot \pi \cdot 0.25}{20 L_0} = \frac{3.927}{L_0} \quad \implies \quad L_0 = \frac{3.927 \sigma}{\eta}. \quad (1)$$

Worked numbers

η	0.02	0.05	0.10	0.30	0.50	0.65
L_0/σ (per compartment)	196	79	39	13.1	7.9	6.0
disks per compartment	50	50	50	50	50	50

So at $\eta = 0.02$ there are still 50 disks on each side; they simply live in a compartment 196 diameters long. Going toward $\eta \rightarrow 0$ along route A means making the box longer, not removing particles.

Why this matters for us

Route A is therefore *not* a density sweep at fixed geometry: density and confinement change together. That is why the $c_s(\eta)$ curve from route A validates the *instrument* (the divider-resonance method), while a *bulk* claim needs a family where N grows at fixed η and fixed aspect ratio (famA), extrapolated in $1/\sqrt{N}$.

2 Kinetic numbers: mean speed, collision rate, mean free path, Knudsen number

2.1 Mean speed in 2D

The 2D Maxwell speed distribution is $f(v) = \frac{m}{k_B T} v e^{-mv^2/2k_B T}$. Then

$$\langle v \rangle = \sqrt{\frac{\pi k_B T}{2m}} = 1.2533, \quad \langle v^2 \rangle = \frac{2k_B T}{m}, \quad \langle |v_x| \rangle = \sqrt{\frac{2k_B T}{\pi m}} = 0.7979, \quad (2)$$

$$\langle v_{\text{rel}} \rangle = \sqrt{2} \langle v \rangle = \sqrt{\frac{\pi k_B T}{m}} = 1.7725 \quad (\text{relative speed of two Maxwellian particles}). \quad (3)$$

2.2 Collision rate per particle

A disk of diameter σ sweeps a “collision strip” of width σ (two centres collide when their distance is σ). In the dilute limit the rate at which one particle collides is number density $\times$ strip width $\times$ mean relative speed; at finite density multiply by the contact value of the pair correlation, $g(\sigma)$:

$$\nu_c = n \sigma g(\sigma) \langle v_{\text{rel}} \rangle = n \sigma g(\sigma) \sqrt{\frac{\pi k_B T}{m}}. \quad (4)$$

For hard disks the equation of state is exactly $Z \equiv \frac{PA}{Nk_B T} = 1 + 2\eta g(\sigma)$, so $g(\sigma) = (Z - 1)/2\eta$.

With $n = 4\eta/(\pi\sigma^2)$:

$$\boxed{\nu_c = \frac{2(Z-1)}{\sqrt{\pi}} \sqrt{\frac{k_B T}{m}} \frac{1}{\sigma}} \xrightarrow{\eta \rightarrow 0} \nu_c \simeq \frac{4\eta}{\sqrt{\pi}}. \quad (5)$$

This is the “physical collision rate” we used to interpret the event budget: at $\eta = 0.69$, $Z \approx 10.1$ gives $\nu_c \approx 10.3$ collisions per particle per σ -time.

2.3 Mean free path

$$\lambda = \frac{\langle v \rangle}{\nu_c} = \frac{1}{\sqrt{2} n \sigma g(\sigma)} \xrightarrow{\eta \rightarrow 0} \lambda = \frac{\pi}{4\sqrt{2}} \frac{\sigma}{\eta} = 0.555 \frac{\sigma}{\eta}. \quad (6)$$

(In the chat I quoted $0.278 \sigma/\eta$; that missed a factor 2. The conclusions below do not change — every Knudsen number just doubles — but use (6).)

2.4 Knudsen number along route A

The Knudsen number compares the mean free path with the container:

$$\text{Kn} = \frac{\lambda}{L_0} = \frac{0.555 \sigma/\eta}{3.927 \sigma/\eta} = 0.141 \quad \text{independent of } \eta \text{ (dilute-limit } g; \text{ at fixed } N = 100). \quad (7)$$

The η cancels because both λ and L_0 scale as $1/\eta$ when N is fixed. Along route A every point has the same ratio “collisions per crossing”, about $1/\text{Kn} \approx 7$ collisions while a particle crosses the compartment.

Worked numbers at the dilute end

At $\eta = 0.0196$ ($L_0 = 200$): $\lambda = 28\sigma$, $\nu_c = 0.046$ per σ -time (using $Z = 1.040$). The hold before release is $2000 \times 0.4/24 = 33.3$ σ -time, i.e. $33.3 \times 0.046 \approx 1.5$ collisions per particle. That is enough only because the velocities are drawn Maxwellian and rescaled to $k_B T = 1$ at $t = 0$; the hold does not *create* equilibrium, it preserves it.

2.5 Reducing the particle number at fixed box: the Knudsen pilot

If instead we keep $L_0 = 200$ and lower the number per side, $\eta = N_{\text{side}}\pi r^2/(L_0 H)$:

Knudsen crossover at fixed box (compartment 200 diameters long)

N_{side}	50	25	10	5
η	0.0196	0.0098	0.0039	0.0020
λ/σ	28	57	141	283
$\text{Kn} = \lambda/L_0$	0.14	0.28	0.71	1.4

Between $N_{\text{side}} = 25$ and 10 the gas stops being a fluid (particles cross the compartment without colliding). This is the regime change of Section 3, and it is why “fewer particles” is a different experiment from “lower density”.

3 The collisionless limit: why $c_s \rightarrow \sqrt{3}$, not $\sqrt{2}$

Take one particle bouncing between a fixed wall and a wall a distance L away that recedes slowly with speed $u = \dot{L}$. Per bounce on the receding wall the particle loses $2u$ of normal speed (rule (19) with $v \rightarrow -v$); it hits that wall once per round trip, every $2L/v_x$, so

$$\frac{dv_x}{dt} = -2u \cdot \frac{v_x}{2L} = -\frac{\dot{L}}{L} v_x \implies \frac{dv_x}{v_x} = -\frac{dL}{L} \implies v_x L = \text{const.} \quad (8)$$

(The adiabatic invariant $\oint p dx = 2mv_x L$ is conserved.) If particles never collide with each other, the y -motion is untouched by a compression along x . The “temperature” of the x -motion is $T_x \propto v_x^2 \propto L^{-2}$ and the pressure on the divider is $P \propto n T_x \propto L^{-1} L^{-2} = L^{-3}$. Compare with an adiabat $P \propto V^{-\gamma}$: this is $\gamma = 3$, the one-dimensional adiabatic index (this is the Lua–Grosberg ideal piston). Then

$$c_s^2 = \gamma \frac{k_B T}{m} : \quad c_s = \sqrt{3} = 1.732 \text{ (collisionless, } \text{Kn} \gg 1) \quad \text{vs} \quad c_s = \sqrt{2} = 1.414 \text{ (fluid, } \text{Kn} \ll 1). \quad (9)$$

The fluid value needs collisions to share the compression energy between x and y ($\gamma = (d+2)/d = 2$ in $d = 2$). Along route A, $\text{Kn} = 0.14$ at every η , so route A sits on the fluid side and the measured dilute value 1.47 at $\eta = 0.02$ is consistent with $\sqrt{2}(1+2\eta) = 1.47$. The Knudsen pilot would show the crossover $1.41 \rightarrow 1.73$ as N_{side} drops — a positive control that the method sees γ , not just a number.

4 Sound speed from the equation of state

4.1 Derivation of $c_s^2 = \frac{k_B T}{m} [Z + \eta Z' + Z^2]$

Hard disks have no potential energy, so $E = Nk_B T$ exactly (2D). Write $P = nk_B T Z(\eta)$ with $n = N/A$, $\eta = n\pi r^2$, so $\eta Z'(\eta) = n dZ/dn$.

Step 1: adiabat. $dE = -P dA$ with $dE = Nk dT$:

$$Nk dT = -\frac{N}{A} k_B T Z dA \implies \frac{dT}{T} = -Z \frac{dA}{A} = Z \frac{dn}{n}. \quad (10)$$

(Ideal gas $Z = 1$: $TA = \text{const}$, i.e. $TV^{\gamma-1}$ with $\gamma = 2$.)

Step 2: isentropic compressibility.

$$\left(\frac{\partial P}{\partial n} \right)_S = k_B T Z + nk Z \left(\frac{\partial T}{\partial n} \right)_S + nk_B T \frac{dZ}{dn} = k_B T [Z + Z \cdot Z + \eta Z']. \quad (11)$$

Step 3: sound speed. Mass density $\rho = mn$, so

$$c_s^2 = \left(\frac{\partial P}{\partial \rho} \right)_S = \frac{k_B T}{m} [Z + \eta Z'(\eta) + Z^2]. \quad (12)$$

4.2 Limits and expansion

- Ideal gas $Z = 1$: $c_s^2 = 2k_B T/m$, $c_s = \sqrt{2} = 1.4142$. (Same as $\gamma k_B T/m$ with $\gamma = 2$.)
- Virial series in 2D: $Z = 1 + 2\eta + 3.128\eta^2 + \dots$ (the $3.128 = 0.782 \times 2^2$ is B_3/B_2^2 times $(2\eta)^2$). Insert in (12):

$$c_s^2 = \frac{k_B T}{m} [2 + 8\eta + 19.6\eta^2 + \dots] \implies c_s = \sqrt{2} (1 + 2\eta + 2.9\eta^2 + \dots). \quad (13)$$

The first-order law $c_s = \sqrt{2}(1 + 2\eta)$ is exact through $O(\eta)$ and good to 1% only for $\eta \lesssim 0.05$.

Kolafa–Rottner values (main reference)

η	0.0196	0.05	0.10	0.30	0.50	0.65
Z_{KR}	1.040	1.109	1.236	2.063	4.106	8.408
c_s from (12)	1.471	1.567	1.744	2.852	5.416	10.44
$\sqrt{2}(1 + 2\eta)$	1.470	1.556	1.697	2.263	—	—

Why this matters for us

Your professor's interest in the ideal-gas limit: (12) is the whole thermodynamics of the fluid in one measurable number, and at $\eta \rightarrow 0$ it collapses to a value with *no free parameter*, $\sqrt{2k_B T/m}$. If the instrument does not return $\sqrt{2}$ there, nothing at higher η can be trusted. The slope 2η is the second virial coefficient — the first real interaction effect — so the dilute end tests both the equilibrium sampling (temperature) and the first term of the EOS. The refit gives $c_s/\sqrt{2} - 1 = 0.035$ at $\eta = 0.0196$ vs $2\eta = 0.039$: the slope is right.

5 How c_s is measured: the divider resonance (Román relation)

5.1 Set-up

Outer wall at $x = 0$ (fixed), gas column of length L , divider of mass M at $x = L$, a second identical column on the other side. Treat the gas as a continuum with sound speed c and mass density ρ ;

small displacement field $u(x, t)$, pressure perturbation $\delta P = -\rho c^2 \partial_x u$ (compression $\partial_x u < 0$ raises P).

5.2 Derivation of $\cot K = \frac{M}{2Nm} K$

Standing wave with the outer wall fixed: $u(x, t) = U \sin(kx) e^{i\omega t}$, $\omega = ck$. The divider displacement is $\xi = u(L)$. When the divider moves right ($\xi > 0$) the left column is stretched and the right one compressed; both push it back:

$$F = H(\delta P_L - \delta P_R) = -2H\rho c^2 k U \cos(kL) e^{i\omega t}. \quad (14)$$

Newton for the divider, $M\ddot{\xi} = F$:

$$-M\omega^2 U \sin(kL) = -2H\rho c^2 k U \cos(kL) \implies M c^2 k^2 \sin(kL) = 2H\rho c^2 k \cos(kL). \quad (15)$$

Multiply by $L/(c^2 k)$ and set $K \equiv kL$; note $H\rho L = Nm$ is the gas mass *per side* ($N = 50$):

$$\boxed{\cot K = \frac{M}{2Nm} K \equiv \alpha K, \quad \nu = \frac{\omega}{2\pi} = \frac{c_s K}{2\pi L_{\text{eff}}}.} \quad (16)$$

$L_{\text{eff}} = L_0 - 2r$: disk centres cannot get closer than r to the outer wall or to the divider, so the acoustic column is shorter than the box by one diameter.

5.3 Two limits as sanity checks

Check

Light divider $M \rightarrow 0$: $\cot K \rightarrow 0$, $K \rightarrow \pi/2$, $\nu = c_s/(4L_{\text{eff}})$ — the quarter-wave resonance of a column open at one end.

Heavy divider $M \gg 2Nm$: K small, $\cot K \approx 1/K$, so $K^2 = 1/\alpha = 2Nm/M$ and $\omega = c_s K/L = \sqrt{2Nm c_s^2/(ML^2)} = \sqrt{2\gamma PH/(LM)} = \sqrt{\kappa/M}$: a mass on two gas springs of stiffness $\kappa/2 = \gamma PH/L$ each (using $c_s^2 = \gamma P/\rho$). This is the textbook adiabatic-piston oscillator.

5.4 Roots for our divider masses ($N = 50$ per side)

Solve $\cot K = \alpha K$ on $(0, \pi/2)$ by bisection ($f(K) = \cot K - \alpha K$ decreases monotonically):

Worked numbers

M/m	1	10	50	100	200	500	1000	2000	5000
$\alpha = M/2Nm$	0.01	0.1	0.5	1	2	5	10	20	50
K	1.555	1.429	1.077	0.860	0.653	0.433	0.311	0.222	0.141
$1/\sqrt{\alpha}$ (heavy approx.)	—	—	1.414	1.000	0.707	0.447	0.316	0.224	0.141
Example: $\eta = 0.10$, $L_0 = 39.3$, $L_{\text{eff}} = 38.3$, $M = 200$: $\nu = 1.744 \times 0.653/(2\pi \times 38.3) = 4.7 \times 10^{-3}$ per σ -time, period ≈ 211 σ -time.									

5.5 How the fit works, and what the intercept tells you

For each η we have 9 masses. Compute $x_M = K(\alpha_M)/(2\pi L_{\text{eff}})$ and measure ν_M (FFT peak of the divider position, averaged over 25 repeats, weight $1/\text{sem}^2$). Equation (16) says $\nu_M = c_s x_M$, a straight line *through the origin* with slope c_s :

$$c_s = \frac{\sum_M w_M x_M \nu_M}{\sum_M w_M x_M^2}, \quad \sigma_{c_s}^2 = \frac{1}{\sum_M w_M x_M^2} \quad (w_M = 1/\text{sem}_M^2). \quad (17)$$

A free-intercept fit $\nu = c_s x + b$ is the diagnostic: b must be zero within error. Where b is significantly non-zero (the refit finds $b \approx -0.001 \dots -0.003$ at $\eta = 0.57\text{--}0.63$, absorbing up to 3.6%) the single-mode continuum picture is being stretched — the compartment is only 6–7 diameters long there.

6 Pressure: the two routes and the extrapolation (Paper 1, briefly)

Pair-virial route. Virial theorem in $d = 2$: $PA = Nk_B T + \frac{1}{d} \langle \sum_i \mathbf{r}_i \cdot \mathbf{F}_i \rangle$. For hard disks the forces are impulses at contact ($|\mathbf{r}_{ij}| = \sigma$, $\Delta \mathbf{p}_i \parallel \hat{\mathbf{r}}_{ij}$), so over an observation time t_{obs}

$$Z_{\text{pair}} = 1 + \frac{m\sigma}{2Nk_B T t_{\text{obs}}} \sum_{\text{collisions}} |\Delta \mathbf{v}_i \cdot \hat{\mathbf{r}}_{ij}|. \quad (18)$$

Wall route. Momentum delivered to the walls: $Z_{\text{wall}} = \frac{A}{Nk_B T} \cdot \frac{1}{t_{\text{obs}} \mathcal{P}} \sum_{\text{wall hits}} 2m|v_n|$ with $\mathcal{P}$ the perimeter. Momentum balance ties the two: in a steady state the momentum flux through any line is the same, so $Z_{\text{pair}} = Z_{\text{wall}}$ up to statistics — that is the internal consistency check in the campaign.

Finite size. Fit $Z(N) = Z_\infty + a/\sqrt{N}$ over $N = 100, 400, 900, 1600$ with weights $1/\sigma_N^2$ and report χ^2 with 2 degrees of freedom; $\chi^2 \gg 4$ means the four sizes do not lie on one $1/\sqrt{N}$ line, i.e. no bulk value can be quoted (this is what happens at $\eta = 0.67$, $\chi_1^2 = 7.6$, and in famA above 0.67). The same procedure is what the bulk c_s claim needs.

7 Paper 2, Level 0: the energy ledger of one piston push

7.1 Piston collision rule

Prescribed-velocity (infinite-mass) piston moving with velocity u ; particle normal velocity v before the hit. In the piston frame the particle reflects elastically, $v - u \rightarrow -(v - u)$:

$$v' = 2u - v, \quad \Delta E = \frac{1}{2}m[(2u - v)^2 - v^2] = 2mu(u - v), \quad \Delta p = m(v' - v) = 2m(u - v). \quad (19)$$

Signs: piston pushing into the gas $u > 0$, particle approaching it $v < 0$ gives $\Delta E = 2mu(u + |v|) > 0$ (energy into the gas); a particle overtaking a receding piston can give $\Delta E < 0$. This is exactly what `resolve_wall` does for `EV_PL/EV_PR` with $M \leq 0$; the code sums ΔE into `work_pistonR`. For a finite-mass piston it applies the elastic two-body rule and books the *piston's* kinetic-energy change — a different quantity, keep them apart.

7.2 First law for the box

With W the summed piston work (on the system), Q the heat through thermal walls (zero for elastic walls):

$$\boxed{W = \Delta \text{KE}_{\text{gas}} + \Delta \text{KE}_{\text{divider}} + \Delta E_{\text{spring}} + Q} \quad \text{residual } R(t) = W - [\dots] \stackrel{!}{=} \text{round-off}. \quad (20)$$

Momentum: $\Delta(\mathbf{p}_{\text{gas}} + \mathbf{p}_{\text{divider}}) = \sum_{\text{piston hits}} \Delta p + \sum_{\text{outer-wall hits}} \Delta p$.

Why this matters for us

This is why Level 0 could not close on the existing runs: the traces log W , E_{spring} , divider and piston velocities, but not KE_{gas} (only in a stdout line every 1000 steps, and those runs were `--quiet`), no per-collision impulses, no gas momentum. So $R(t)$ has one unrecorded term. The fix is a logging change, not physics: a per-sample KE_{gas} (total and per compartment) column and a gated per-event piston log $(t, u, v, v', \Delta E, \Delta p)$. And the reference runs used `--edmd-acc=1`, the backend that failed validation; they must be repeated on `--edmd-acc=0` before any number from them is physics.

8 Quasi-static work and the definition of dissipation

8.1 Adiabatic compression of hard disks

Compress one compartment from area A_i to A_f (packing $\eta_i \rightarrow \eta_f$) with elastic walls and no heat exchange. Integrate the adiabat (10), $dT/T = Z d\eta/\eta$:

$$\frac{T_f}{T_i} = \exp \left[\int_{\eta_i}^{\eta_f} \frac{Z(\eta)}{\eta} d\eta \right], \quad W_{\text{qs}} = \Delta E = Nk(T_f - T_i) = Nk_B T_i \left(\frac{T_f}{T_i} - 1 \right). \quad (21)$$

Ideal gas ($Z = 1$): $T_f/T_i = \eta_f/\eta_i = A_i/A_f$, so

$$W_{\text{qs}}^{\text{ideal}} = Nk_B T_i \left(\frac{A_i}{A_f} - 1 \right) \quad (\gamma = 2). \quad (22)$$

Worked numbers

$N = 50$ per side, $k_B T_i = 1$, $\eta_i = 0.10$: compressing by 10% ($\eta_f = 0.11$) gives $T_f/T_i = 1.127$ with KR (ideal: 1.100), so $W_{\text{qs}} = 50 \times 0.127 = 6.3$ (ideal 5.0). A 25% compression ($\eta_f = 0.125$) gives $W_{\text{qs}} = 16.4$ (ideal 12.5). At $\eta_i = 0.30$, a 10% compression already costs $W_{\text{qs}} = 11.4$ (ideal 5.0) — the EOS matters even for “small” pushes.

8.2 Isothermal free energy (needed for Jarzynski)

From $dF = -P dA$ at fixed T and $dA/A = -d\eta/\eta$:

$$\Delta F = Nk_B T \int_{\eta_i}^{\eta_f} \frac{Z(\eta)}{\eta} d\eta \quad (\text{ideal: } Nk_B T \ln \frac{A_i}{A_f}). \quad (23)$$

Note $\Delta F \neq W_{\text{qs}}$: the adiabatic quasi-static work heats the gas, the isothermal one does not. For $\eta = 0.10 \rightarrow 0.11$, $N = 50$: $\Delta F = 50 \times 0.119 = 6.0$.

8.3 Dissipation

For a push at finite speed the work fluctuates from run to run. Define

$$W_{\text{diss}} \equiv \langle W \rangle - W_{\text{qs}} \geq 0 \quad (\text{adiabatic, elastic walls: minimum-work principle}), \quad (24)$$

and, for a gas prepared canonically at T , the second-law form $\langle W \rangle \geq \Delta F$ (next section). W_{diss} is the energy that went into the gas beyond what a reversible compression would have put there; after the divider ring-down it is extra heat in the gas. It is the first “lost” quantity we can measure with the ledger closed.

9 Jarzynski: turning fluctuations into a free-energy check

9.1 Statement and the one-line proof for a Hamiltonian system

Start each trajectory from the canonical distribution $\rho_i(\Gamma) = e^{-\beta H_i(\Gamma)} / Z_i$ (draw Maxwellian velocities at T and an equilibrated configuration — no thermal wall needed during the push). Drive the piston with the same protocol; the dynamics are Hamiltonian, so phase-space volume is conserved (Liouville) and $W = H_f(\Gamma_f) - H_i(\Gamma_i)$:

$$\langle e^{-\beta W} \rangle = \int d\Gamma_i \frac{e^{-\beta H_i(\Gamma_i)}}{Z_i} e^{-\beta [H_f(\Gamma_f) - H_i(\Gamma_i)]} = \frac{1}{Z_i} \int d\Gamma_f e^{-\beta H_f(\Gamma_f)} = \frac{Z_f}{Z_i} = e^{-\beta \Delta F}. \quad (25)$$

Jensen (e^{-x} convex): $e^{-\beta \langle W \rangle} \leq \langle e^{-\beta W} \rangle$, hence $\langle W \rangle \geq \Delta F$.

9.2 Why the piston speed must be small

The estimator $\Delta F_{\text{est}} = -k_B T \ln \frac{1}{N} \sum_k e^{-\beta W_k}$ is dominated by the rare low- W trajectories. The rule of thumb: you need of order $e^{\beta W_{\text{diss}}}$ trajectories to see them.

Worked numbers

$\beta W_{\text{diss}} = 1 \rightarrow 3$ trajectories; $3 \rightarrow 20$; $5 \rightarrow 150$; $10 \rightarrow 2 \times 10^4$. With 10^3 – 10^4 trajectories per protocol we can afford $W_{\text{diss}} \lesssim 5$ – $8 k_B T$. The existing reference runs ($N = 600$, work target $2 \times 10^4 k_B T$, piston travelling 10σ in about one σ -time, i.e. u/c_s of order 2–3) are shocks: βW_{diss} in the thousands, Jarzynski useless there. This fixes the design: $N = 50$ – 300 per side, $u/c_s = 0.01$ – 0.3 , compression 10–25%.

9.3 Fast-piston limit (Lua–Grosberg) as an exact reference

If the push is so fast that each particle hits the piston at most once ($u \tau_{\text{push}} \ll L$ and $\text{Kn} \gg 1$), the work is a sum of independent single-hit terms (19) over the particles the piston catches, and $P(W)$ follows from the Maxwell distribution alone. That gives closed-form $\langle W \rangle$, $\text{Var } W$ and $P(W)$ to test the work histograms against at one end of the speed range, while the quasi-static formula (21) anchors the other end.

10 Linear response: dissipation from equilibrium fluctuations

10.1 Friction from the force autocorrelation

Hold the divider fixed at position λ in an equilibrium run and record the net force $F(t)$ on it (sum of impulses from both sides per sampling interval). Linear response (Sivak–Crooks) says that a slow protocol $\lambda(t)$ dissipates

$$W_{\text{diss}} \simeq \int \zeta(\lambda) \dot{\lambda}^2 dt, \quad \zeta(\lambda) = \beta \int_0^\infty \langle \delta F(0) \delta F(t) \rangle_\lambda dt, \quad (26)$$

with $\delta F = F - \langle F \rangle$. So a *fixed*-divider equilibrium run predicts the dissipation of a *moving*-divider experiment. That is the check: measure ζ from fluctuations, then measure W_{diss} directly from the ledger at several u , and confirm $W_{\text{diss}} \propto u^2 \tau = u \Delta x$ at small u .

10.2 A collisionless estimate of ζ (order of magnitude)

For a wall moving at speed u into a gas whose particles do not collide with each other, the pressure it feels is the momentum flux of the particles it catches:

$$P(u) = 2mn \int_{-\infty}^u (u - v_x)^2 f(v_x) dv_x, \quad P(0) = nk_B T, \quad \left. \frac{dP}{du} \right|_0 = 2mn \langle |v_x| \rangle = 2n \sqrt{\frac{2mk_B T}{\pi}}. \quad (27)$$

Two sides, height H : $\zeta \approx 4nH \sqrt{2mk_B T/\pi}$. The dissipated work for a displacement Δx at speed u is $\zeta u \Delta x$, and the quasi-static work is $\approx nk_B T H \Delta x$, so

$$\frac{W_{\text{diss}}}{W_{\text{qs}}} \approx 4 \sqrt{\frac{2}{\pi}} \frac{u}{\sqrt{k_B T/m}} = 3.2 \frac{u}{\sqrt{k_B T/m}} \approx 4.5 \frac{u}{c_s} \quad (\text{dilute}). \quad (28)$$

Worked numbers

$u/c_s = 0.01 \rightarrow 4.5\%$ dissipation; $0.1 \rightarrow 45\%$; $0.3 \rightarrow$ outside linear response. In the collisional (fluid) regime the true friction is smaller (collisions let the gas re-equilibrate ahead of the wall) and (28) is an upper-side estimate; the measured ζ from (26) replaces it. Either way, $u/c_s \approx 0.01$ – 0.1 is where W_{diss} is a few percent to tens of percent of W_{qs} : measurable, and small enough in $k_B T$ for Jarzynski once W_{qs} is tens of $k_B T$.

11 Nostalgia: the information that is lost

11.1 Definitions

Let x_t be the *signal* (the piston protocol, drawn at random from a stochastic rule so that it has entropy) and s_t the *system state* (gas + divider, coarse-grained, e.g. divider position and velocity binned). Mutual information $I(X; Y) = H(X) - H(X|Y) = \sum p(x, y) \ln \frac{p(x, y)}{p(x)p(y)} \geq 0$. Define

$$I_{\text{mem}} = I(s_t; x_t), \quad I_{\text{pred}} = I(s_t; x_{t+1}), \quad \boxed{I_{\text{nost}} = I_{\text{mem}} - I_{\text{pred}} \geq 0}. \quad (29)$$

Memory is what the system's state tells you about the signal now; prediction is what it tells you about the signal next; nostalgia is the part of the memory that is useless for the future.

11.2 Why $I_{\text{nost}} \geq 0$: the data-processing inequality

If the signal is Markov and the system at time t has only seen the signal up to t , then x_{t+1} depends on s_t only through x_t : the chain $s_t \rightarrow x_t \rightarrow x_{t+1}$ is Markov, $p(x_{t+1}|x_t, s_t) = p(x_{t+1}|x_t)$. Then, using the chain rule twice,

$$I(s_t; x_t, x_{t+1}) = I(s_t; x_t) + \underbrace{I(s_t; x_{t+1}|x_t)}_{=0 \text{ (Markov)}} \quad (30)$$

$$= I(s_t; x_{t+1}) + \underbrace{I(s_t; x_t|x_{t+1})}_{\geq 0} \implies I(s_t; x_t) \geq I(s_t; x_{t+1}). \quad (31)$$

The one-variable version is the inequality in your screenshot: if $Y = f(Z)$ is a (possibly coarse-graining) function of Z , then $H(Y) \leq H(Z)$ and $I(Y; X) \leq I(Z; X)$ — processing can only destroy information. The proof is the same chain-rule step: $H(Z) = H(Z, Y) = H(Y) + H(Z|Y) \geq H(Y)$ because $H(Z, f(Z)) = H(Z)$ and conditional entropy is non-negative.

11.3 The bound that connects it to the ledger (Still, Sivak, Bell, Crooks 2012)

For a system driven by a signal, with $W_{\text{diss}}[x_t \rightarrow x_{t+1}]$ the dissipation during one signal step,

$$\boxed{\beta \langle W_{\text{diss}}[x_t \rightarrow x_{t+1}] \rangle \geq I_{\text{mem}} - I_{\text{pred}} = I_{\text{nost}}.} \quad (32)$$

Reading: every nat of useless memory the system keeps about the driving costs at least $k_{\text{B}}T$ of dissipation. The left side comes from the closed ledger of Section 7 (this is why Level 0 must close first); the right side comes from the joint histograms $p(s_t, x_t)$ and $p(s_t, x_{t+1})$ over many trajectories. Both sides are measurable in the same runs.

11.4 What the runs must provide

- A stochastic protocol: e.g. $x_t \in \{\text{push, hold, pull}\}$ or a discretised piston velocity drawn from a Markov chain with known transition matrix, one draw per interval Δt (a few divider periods). Deterministic protocols have $H(x) = 0$ and nothing to compute.
- Per-trajectory logs of x_t , the coarse-grained s_t , and the work per interval (from the per-event piston log, summed over the interval).
- Enough trajectories for the plug-in estimator. Its bias is about $\frac{(k_s - 1)(k_x - 1)}{2N_{\text{traj}}}$ nats for k_s state bins and k_x signal bins: with $k_s = 10$, $k_x = 3$ and $N_{\text{traj}} = 10^3$ the bias is 0.009 nats; with 10^4 it is 0.001. That is the scale on which I_{nost} must sit above zero to mean anything, and it sets the 10^3 – 10^4 trajectory count.
- Small N (50–300 per side) and $u/c_s \leq 0.3$, so that βW_{diss} per interval is of order one — then (32) is a tight, testable statement rather than $10^3 \geq 0.01$.

12 The chain, in one table

Level	Quantity	Reference to compare with
Paper 1	$Z(\eta)$, $c_s(\eta)$	KR EOS, (12); ideal limit $\sqrt{2}$; famA $1/\sqrt{N}$ extrapolation
0	ledger residual $R(t)$, momentum	round-off (elastic walls)
1	divider fluctuations, ζ	equipartition $\frac{1}{2}M \langle \dot{\xi}^2 \rangle = \frac{1}{2}k_{\text{B}}T$; (26)
2	$P(W)$ at several u	(21) slow end, Lua–Grosberg fast end, Jarzynski (25)
3	$W_{\text{diss}}(u)$	$\propto u$ at small u , prefactor ζ from Level 1
4	$I_{\text{mem}}, I_{\text{pred}}, I_{\text{nost}}$	data processing ≥ 0 ; bound (32) against Level 3